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Department of Mathematics
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Stat 201 Major Exam I Term 243

Original exam date: 3 July 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes Original course: STAT 319

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1. There is a chance that a bit transmitted through a digital transmission channel is received in error. Let X equal the number of bits in error in the next four bits transmitted. Assume that the probability mass function of X is given by

$$P(X = x) = \begin{cases} ax & x = 1, 2, 3 \text{ or } 4 \\ 0 & \text{otherwise} \end{cases}$$

The expected value of the number of properly functioning components

(a) 2

(b) 1

(c) 5

(d) 3

(e) 4

2. The data are available on compressive strengths in pounds per square inch (psi) of 80 specimens of a new aluminum-lithium alloy undergoing evaluation as a possible material for aircraft structural elements. A summary descriptive statistic of the data is given below:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
76	144.5	161.5	162.7	181	245

Which of the following is true?

(a) There are extreme outliers in the data

(b) There are no outliers in the data

(c) The interquartile range equal to 36.5

(d) The data negatively skewed (left skewed)

(e) The data symmetric around the mean

3. A batch contains 40 bacterial cells, 12 of which are unable to replicate. If three cells are randomly selected without replacement, what is the probability that at least one of them is unable to replicate?

(a) 0.2049

(b) 0.3316

(c) 0.7951

(d) 0.6684

(e) 0.6830

4. In a sample of 30 heart failure patients, each independently having a 13% chance of being affected by external factors, what is the probability that at least 3 patients are affected by external factors?

(a) 0.130

(b) 0.870

(c) 0.767

(d) 0.792

(e) 0.208

5. Which of the following statements is NOT true?

(a) The symmetry implies the mode is at the median of the sample

(b) The median can be greater than the mean when the sample is symmetric about the mean value

(c) The median will be equal to the mean when the sample is symmetric about the mean value

(d) The mean is sensitive to extreme values while the median is not sensitive to the extreme values

(e) The median will equal the mode when the sample is symmetric with a single mode

6. Samples of emissions from three suppliers are classified for conformance to air-quality specifications. The results from 100 samples are summarized as follows:

	Conforms	
Supplier	Yes	No
H	22	8
L	25	5
M	30	10

Let A denote the event that a sample is from supplier H, and let B denote the event that a sample conforms to specifications. If a sample is selected at random, then $P(\overline{A} \mid \overline{B}) =$

(a) 0.6522

(b) 0.1139

(c) 0.7619

(d) 0.4286

(e) 0.2025

7. If A, B, and C are mutually exclusive events with $P(A) = 0.2$, $P(B) = 0.3$, and $P(C) = 0.4$, then $P(\overline{A} \cap \overline{B} \cap \overline{C}) =$

(a) 0.9

(b) 0.8

(c) 0.2

(d) 0.1

(e) 0

8. The following data are direct solar intensity measurements ($watts/m^2$) on different days at a location in southern Spain:

562	869	708	775	775	704	809	856	655	806	878	909
918	558	768	870	918	940	946	661	820	898	935	952
957	693	835	905	939	955	960	498	653	730	753	

Given that $\sum_{i=1}^{35} x_i = 28368$ and $\sum_{i=1}^{35} x_i^2 = 23552500$. The estimated value of the population standard deviation.

(a) 810.5143

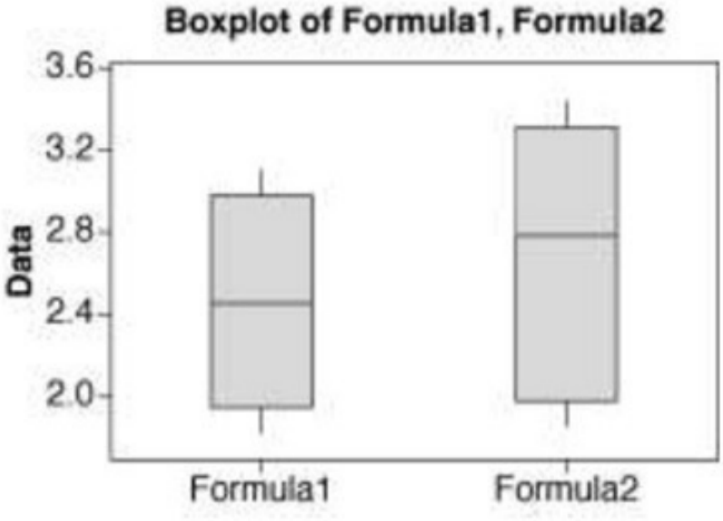
(b) 126.4719

(c) 15995.16

(d) 16465.61

(e) 128.3184

9. The "cold start ignition time" of an automobile engine is being investigated by a gasoline manufacturer. Two formulation of the gasoline was tested in the same vehicle. Use the following graph to comment on both formulas (box plots of Formula1 and Formula2). Which of the following statements is true?



(a) Formula 2 shows more consistent performance than Formula 1

(b) Formula 1 has a lower mean cold start ignition time and is more consistent

(c) Both formulas have identical performance characteristics

(d) Formula 1 has more variation in ignition time than Formula 2

(e) Formula 2 has a lower median ignition time than Formula 1, indicating better performance

10. The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 8 calls per hour, what is the probability that there are exactly 5 calls in 30 minutes?

(a) 0.1843

(b) 0.0992

(c) 0.0424

(d) 0.0916

(e) 0.1563

11. An email filter is being developed to help distinguish between spam and real (non-spam) emails. In this system, 10% of all emails are spam. The word 'free' appears in 80% of spam emails and in only 3% of real emails. What is the probability that a randomly selected email does not contain the word 'free'?

(a) 0.252

(b) 0.107

(c) 0.893

(d) 0.089

(e) 0.978

12. A lot of 50 spacing washers contains 30 washers that are thicker than the target dimension. Washers are selected from the lot at random without replacement; what is the minimum number of washers that need to be selected so that the probability that all the washers are thinner than the target is less than 0.10?

(a) 2

(b) 1

(c) 5

(d) 3

(e) 4

13. Which of the following statements is true?

(a) The sample standard deviation can be equal to zero

(b) Suppose that you add 10 to all of the observations in a sample, the sample standard deviation will change

(c) The sample mean will always correspond to one of the observations in the sample

(d) Exactly half of the observations in a sample will fall below the mean

(e) The sample mean will always be the most frequently occurring data value in the sample

14. The thickness of wood paneling (in inches) that a customer orders is a random variable with the following cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.2 & 1 \leq x < 2 \\ 0.5 & 2 \leq x < 3 \\ 0.9 & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

What is the probability that a randomly selected order is for paneling at least 3 inches thick?

(a) 0.5

(b) 0.1

(c) 0.7

(d) 0.2

(e) 0.3

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$a = 0.1$, $E(X) = 3$ errors, so $E(4 - X) = 1$.
2	**(c)**	IQR $= 181 - 144.5 = 36.5$; outliers are mild only.
3	**(d)**	$1 - \binom{28}{3}/\binom{40}{3} = 0.6684$.
4	**(c)**	$1 - P(X \le 2)$ with $X \sim \text{Bin}(30, 0.13)$ gives 0.767.
5	**(b)**	Symmetric means median = mean, so 'median can exceed the mean' is false.
6	**(a)**	$P(\bar{A} \mid \bar{B}) = 15/23 = 0.6522$.
7	**(d)**	$1 - (0.2 + 0.3 + 0.4) = 0.1$.
8	**(e)**	$s^2 = 16465.61$, so $s = 128.3184$.
9	**(b)**	Formula 1: lower centre, shorter box, so less variation.
10	**(e)**	$\lambda = 4$; $P(5) = e^{-4}4^5/5! = 0.1563$.
11	**(c)**	$P(\text{free}) = 0.107$, so the answer is $1 - 0.107 = 0.893$.
12	**(d)**	$P = \frac{20}{50}\frac{19}{49}\frac{18}{48} = 0.058 < 0.10$ first at $n = 3$.
13	**(a)**	$s = 0$ when all values are equal; the others are false.
14	**(a)**	$P(X \ge 3) = 1 - F(2) = 0.5$.